

SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Code of Conduct:

I M. Farah Bawani/19251434 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Farah Bawani
Signature of the student:

Criteria	Weightage (%)	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks	QS 10marks
Understanding of Scenario	20%	1.4	1.4	1.6	1.6	1
Identification of Key Issues	20%	1.6	1.6	1.4	1.4	1
Application of Concepts	25%	1	1	1	1	1.6
Decision Making	20%	1	1	1	1	1
Clarity of Response	15%	1	1	1	1	0.4

28/3/20

M. Farah Abrin
192511434

Co-2

Assessment Tool-2

Scenario-based

Given:

Network Block: 192.168.100.0/24

Departments and subnet masks:

↳ School of computing → /25

↳ School of Electronics → /26

↳ School of mechanical Engineering → /27

i) School of computing (125)

* Subnet mask: 255.255.255.128

* Total IP addresses = $2^{32-25} = 128$

* Usable hosts = $128 - 2 = 126$

* Network Address = 192.168.100.0

* Valid host range = 192.168.100.1 - 192.168.100.126

* Broadcast Address = 192.168.100.127

ii) School of electronics (126)

Next available subnet begins at 192.168.100.128

* Subnet mask: 255.255.255.192

* Total IP addresses: 64

* Usable hosts: $64 - 2 = 62$

* Network Address: 192.168.100.128

* Valid host range : $192.168.100.129 - 192.168.100.190$

* Broadcast Address : $192.168.100.191$

iii) School of mechanical Engineering (127)

* Subnet mask : $255.255.255.224$

* Total IP addresses : 32

* Usable Hosts = $32 - 2 = 30$

* Network Address : $192.168.100.192$

* Valid host range : $192.168.100.193 - 192.168.100.222$

* Broadcast Address : $192.168.100.223$

iv) Remaining unused IP Address Range

Original Network :

* $192.168.100.0 - 192.168.100.255$

Allocated ~~Network~~ Range :

* $192.168.100.0 - 192.168.100.223$

Unused IP Address Range :

* $192.168.100.224 - 192.168.100.255$

The remaining block contains 32 IP addresses, which can be used for future subnet allocation

network Block: 10.10.0.0/16
Subnet Allocation:

* Software development $\rightarrow$ /24

* Quality Assurance $\rightarrow$ /25

* Human Resources $\rightarrow$ /26

* Executive Management $\rightarrow$ /27

i) Software Development (124)

* Network Address: 10.10.0.0/24

* Subnet Mask: 255.255.255.0

* Broadcast Address: 10.10.0.255

* Host Range: 10.10.0.1 - 10.10.0.254

* Usable hosts: 254

ii) Quality Assurance (125)

Next available subnet starts at 10.10.1.0

* Network Address: 10.10.1.0/25

* Subnet Mask: 255.255.255.128

* Broadcast Address: 10.10.1.127

* Host Range: 10.10.1.1 - 10.10.1.126

* Usable Host: 125

iii) Human resources (126)

next available subnet starts at 10.10.1.128

* Network Address :

* Subnet Mask :

* Broadcast Address :

* Host Range

* Usable hosts

iv) Executive Management (127)

next available subnet starts at 10.10.1.192

* Network Address : 10.10.1.192/27

* Subnet Mask : 255.255.255.224

* Broadcast Address : 10.10.1.223

* Host Range : 10.10.1.193 - 10.10.1.222

* Usable hosts : 30

Remaining unused IP Address Range

→ Remaining in the current second octet

■ 10.10.1.224 - 10.10.1.255 (32 IP addresses)

→ Remaining in the 1/16 block

All addresses from:

■ 10.10.2.0 - 10.10.255.255

remain available for future subnetting

IPv6 Address:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

i) Convert the IPv6 Address into Compressed notation

Rules — $\left\{ \begin{array}{l} \rightarrow \text{Remove leading zeros in each block} \\ \rightarrow \text{Replace consecutive blocks of 0000 with ::} \end{array} \right.$

Compressed Address:

2001:db8:abcd::/64

ii) Expand the Compressed Address back to Full form

Expanding:

2001:db8:abcd::/64



2001:0db8:abcd:0000:0000:0000:0000:0000/64

iii) Expand the compressed Address back to Full form

Expanding

2001:db8:abcd::/64

Full IPv6 Address:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

iv) Identify the Network Prefix using the /64

Prefix length

* Prefix length : 64

* Network Prefix (64 bits): 2001:0db8:abcd:0000::/64

* Network Portion : 2001:0db8:abcd:0000

* Host Portion : 0000:0000:0000:0000

iv) Calculate the total number of host addresses supported

A /64 IPv6 network leaves

- Total bits = 128

- Network bits = 64

- Host bits = 64

Total host addresses : 2^{64}

$$= 18,446,744,073,709,551,616$$

v) Total number of possible host addresses available:

Since IPv6 does not reserve broadcast or network addresses like IPv4:

$$18,446,744,073,709,551,616 (2^{64})$$

Routing Table

Destination network	Prefix
192.168.1.0	124
192.168.2.0	124
192.168.3.0	124

Destination IP Address : 192.168.2.45

i) Determine the destination subnet

A 124 subnet mask is 255.255.255.0

Network ID of 192.168.2.45:

IP Address : 192.168.2.45

Subnet mask : ~~255.255.255.0~~

Network ID : 192.168.2.0

Destination Subnet : 192.168.2.0/24

ii) Matching Routing Table Entry

Routing Entry	Match
192.168.1.0/24	NO
192.168.2.0/24	Yes
192.168.3.0/24	NO

iii) Next-Hop Network / Interface

The router forwards the packet through the (or next hop) associated with the matching next-hop network / interface. Interface connected 192.168.2.0/24 (e.g., interface 60/1)

iv) Default route

0.0.0.0/0

⑤ Given:

<u>Department</u>	<u>Subnet</u>
Administration LAN	192.168.10.64/26
Finance LAN	192.168.10.64/26

Router Interfaces (Gateways):

- Administration Gateway → 192.168.10.1
- Finance Gateway → 192.168.10.65

Network Address and Broadcast address

Administration LAN (192.168.10.0/26)

* Network Address: 192.168.10.0

* Broadcast Address: 192.168.10.63

Carve LAN (192.168.10.64/26)

Network Address

* Broadcast Address

ii) Valid Host IP Address Range

Administration LAN

First Host : 192.168.10.1

Last Host : 192.168.10.62

Host Range:

192.168.10.1 - 192.168.10.62

Finance Range LAN:

* First Host : 192.168.10.65

* Last Host : 192.168.10.126

Host Range : 192.168.10.65 - 192.168.10.126

iii) Suitable IP Addresses for three computers

Administration LAN

Computer	IP Address
Pc1	192.168.10.2
Pc2	192.168.10.3
Pc3	192.168.10.4

Finance LAN

Computer	IP address
Pc1	192.168.10.66
Pc2	192.168.10.67
Pc3	192.168.10.68

iv) Default Gateway

LAN	Default Gateway
Administration LAN	192.168.10.1
Finance LAN	192.168.10.65